Theory of Elementary Particles
Lecture 10

Krzysztof M. Graczyk
kgraczyk.github.io

Institute of Theoretical Physics
University of Wrocław
pl. M. Borna 9, 50-204 Wrocław, Poland

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Goldstone's Theorem
- Functional attempt
- Symmetry Breaking and Operators

Weak interactions: Fermi Model
- Left-Right handed neutrinos
- Problems of Fermi Model

Symmetry $SU(2) \times U(1)$ for electroweak interactions

Literature

- Stefan Scherer and Matthias R. Schindler, *A Chiral Perturbation Theory Primer*, arXiv:hep-ph/0505265
- Howard Georgi , *Weak Interactions and Modern Particle Theory*, Benjamin-Cummings Pub Co, 1984.

Goldstone's Theorem

Why do we need symmetry breaking?

Symmetry breaking leads to occurrence of scalar particles (Goldstone bosons): it refers to global symmetry and its breaking,

Functional Attempt

- Consider the symmetry Lie (compact) group G of order n_G i.e. the algebra is generated by n_G generators
- Let us consider the Lagrangian

$$\mathcal{L}(\vec{\Phi}, \partial_\mu \vec{\Phi}) = \frac{1}{2} \partial_\mu \vec{\Phi} \cdot \partial^\mu \vec{\Phi} - \mathcal{V}(\vec{\Phi}), \tag{1}$$

  where $\vec{\Phi}$ is multiplet of scalar (or pseudoscalar) Hermitian fields
- Assume that the above Lagrangian is invariant under G, where the infinitesimal transformations of the fields are given by

$$g \in G: \quad \Phi_k \to \Phi_k + \delta\Phi_k, \quad \delta\Phi_k = -i\epsilon_a T_{a,kj} \Phi_j \tag{2}$$

  and $(T_a)_{ij}$ is a hermitian representation, so that it is antisymmetric and purely imaginary

* In the case of $SU(2)$ it is adjoint representation.

- Assume that $\mathcal{V}(\vec{\Phi})$ has a global minimum and it may generate spontaneous symmetry breaking in a ground state with a vacuum expectation value

$$\vec{\Phi}_{min} = \langle \vec{\Phi} \rangle \tag{3}$$

- Assume that the choice of the vacuum is invariant under a continuous subgroup H of G

Functional Attempt

- Let us expand $\mathcal{V}$ about $\vec{\Phi}_{min}$, $\mid \vec{\Phi}_{min} \mid = \nu$, i.e.

$$\vec{\Phi} = \vec{\Phi}_{min} + \vec{\chi} \tag{4}$$

Then

$$\mathcal{V}(\vec{\Phi}) = \mathcal{V}(\Phi_{min}) + \frac{\partial \mathcal{V}(\Phi_{min})}{\partial \Phi_k}\chi_k + \frac{1}{2}\frac{\partial^2 \mathcal{V}(\Phi_{min})}{\partial \Phi_k \partial \Phi_j}\chi_k \chi_j + \ldots \tag{5}$$

- At the minimum

$$\frac{\partial \mathcal{V}(\Phi_{min})}{\partial \Phi_k} = 0 \tag{6}$$

while symmetric matrix

$$M^2 \equiv m^2_{kj} = \frac{\partial^2 \mathcal{V}(\Phi_{min})}{\partial \Phi_k \partial \Phi_j} \tag{7}$$

must be positive semi-definite i.e.

$$\sum_{k,j} m^2_{kj} x_k x_j \geq 0 \tag{8}$$

for every $\vec{x}$, which means that all eigenvalues of M^2 are non-negative!

Functional Attempt

- Let us transform the fields, according to the symmetry transformation, so that

$$\vec{\Phi} \rightarrow \vec{\Phi} + \delta\vec{\Phi} \tag{9}$$

Then

$$\begin{aligned} \mathcal{V}(\vec{\Phi}_{min}) &\rightarrow \mathcal{V}(\vec{\Phi}_{min} + \delta\vec{\Phi}_{min}) && (10) \\ &\rightarrow \mathcal{V}(\Phi_{min}) + \frac{1}{2} m^2_{kj} \delta\Phi_{min,k} \delta\Phi_{min,j} + \ldots && (11) \end{aligned}$$

- so

$$m^2_{kj} \delta\Phi_{min,k} \delta\Phi_{min,j} = 0 \tag{12}$$

- Differentiating with respect to $\delta\Phi_n$ and using $m^2_{kj} = m^2_{jk}$ leads to

$$M^2 \delta\vec{\Phi}_{min} = 0 \tag{13}$$

or

$$M^2 T_a \vec{\Phi}_{min} = 0 \tag{14}$$

Functional Attempt

1. T_a, $a = 1, ..., n_H$ is a representation of an element of the Lie algebra belonging to the subgroup H of G leaving the selected ground state invariant hence

$$T_a \vec{\Phi}_{min} = \vec{0}, \quad a = 1, ..., n_H \tag{15}$$

because

$$\exp(-i\epsilon_a T_a)\Phi_{min} = \Phi_{min} \Rightarrow (1 - i\epsilon_a T_a)\Phi_{min} \approx \Phi_{min}$$

2. T_a, $a = n_H + 1, ..., n_G$ is not a representation of an element of the Lie algebra belonging to the subgroup H. Hence

$$T_a \vec{\Phi}_{min} \neq \vec{0}, \quad a = n_H + 1, ..., n_G \tag{16}$$

which means that $\delta\vec{\Phi}_{min}$ **is eigenvector of** M^2 with eigenvalue 0!

3. It is easy to show that $T_a\Phi_{min}$ for different as are linearly independent
4. $T_a\Phi_{min}$, $a = n_H + 1, ..., n_G$ are massless Goldstone bosons. There are $n_G - n_H$ such bosons.

Symmetry Breaking and Operators: example SO(3)

- Consider a Hamiltonian with global symmetry group $G = SO(3)$, let

$$\vec{\Phi} = \begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \end{pmatrix} \tag{17}$$

denotes a triplet of Hermitian operators transforming as a vector under G

$$g \in G: \quad \vec{\Phi}(x) \to \vec{\Phi}'(x) = \exp[i\alpha_k Q_k]\vec{\Phi}(x)\exp[-i\alpha_k Q_k] = \exp[-i\alpha_k T_k]\vec{\Phi}(x) \tag{18}$$

while Q_is are generators of the $SO(3)$ transformations in the Hilbert space and they satisfy

$$[Q_n, Q_j] = i\varepsilon_{njk}Q_k \tag{19}$$

while $T_{k,nj}$ are the matrices of the three dimensional representation of $SO(3)$ so that $T_{k,nj} = -i\varepsilon_{knj}$

Symmetry Breaking and Operators: example SO(3)

Now assume that

$$0 = \langle 0 \mid \Phi_1(x) \mid 0 \rangle = \langle 0 \mid \Phi_2(x) \mid 0 \rangle \quad (20)$$

$$0 \neq \nu = \langle 0 \mid \Phi_3(x) \mid 0 \rangle \quad (21)$$

Then the symmetry breaks i.e. $SO(3) \to SO(2)$

Theorem (version of the Goldstone Theorem)

Q_1 and Q_2 do not annihilate the ground state and to each such generator corresponds a massless Goldstone boson.

$$\vec{\Phi}' \approx \vec{\Phi} + i\alpha_k Q_k \vec{\Phi} - i\vec{\Phi}\alpha_k Q_k = \vec{\Phi} + i\alpha_k [Q_k, \vec{\Phi}] \tag{22}$$

on the other hand

$$\vec{\Phi}' = (1 - i\alpha_k T_k)\vec{\Phi} = \vec{\Phi} + \vec{\alpha} \times \vec{\Phi} \tag{23}$$

hence one gets

$$i\alpha_k [Q_k, \Phi_j] = \varepsilon_{jkm} \alpha_k \Phi_m \tag{24}$$

which works for any choice of $\vec{\alpha}$, hence

$$[Q_k, \Phi_j] = -i\varepsilon_{jkm}\Phi_m = i\varepsilon_{kjm}\Phi_m \tag{25}$$

Notice that

$$\varepsilon_{kjm}\varepsilon_{kjn} = 2\delta_{nm} \tag{26}$$

then

$$\varepsilon_{kjn}[Q_k, \Phi_j] = i\varepsilon_{kjn}\varepsilon_{kjm}\Phi_m \tag{27}$$

$$\varepsilon_{kjn}[Q_k, \Phi_j] = 2i\Phi_n \tag{28}$$

In particular

$$\Phi_3 = -\frac{i}{2}\left([Q_1, \Phi_2] - [Q_2, \Phi_1]\right) \tag{29}$$

- Consider the rotation around second axis, namely,

$$e^{-i\frac{\pi}{2}T_2}\vec{\Phi} = \begin{pmatrix} \cos(\pi/2) & 0 & \sin(\pi/2) \\ 0 & 1 & 0 \\ -\sin(\pi/2) & 0 & \cos(\pi/2) \end{pmatrix} \vec{\Phi} = \begin{pmatrix} \Phi_3 \\ \Phi_2 \\ -\Phi_1 \end{pmatrix} = e^{i\frac{\pi}{2}Q_2} \begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \end{pmatrix} e^{-i\frac{\pi}{2}Q_2} \tag{30}$$

  One gets the equations

$$\Phi_1 = -e^{i\frac{\pi}{2}Q_2}\Phi_3 e^{-i\frac{\pi}{2}Q_2}, \quad \Phi_3 = e^{i\frac{\pi}{2}Q_2}\Phi_1 e^{-i\frac{\pi}{2}Q_2} \tag{31}$$

  Indeed

$$e^{i\frac{\pi}{2}Q_2}\Phi_3 e^{-i\frac{\pi}{2}Q_2} = \Phi_3 + i\frac{\pi}{2}[Q_2, \Phi_3] + \cdots = \Phi_3\cos(\pi/2) - \Phi_1\sin(\pi/2) = -\Phi_1 \tag{32}$$

Baker-Campbell-Hausdorff formula

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2}[A, [A, B]] + ... + \frac{1}{n!}[A, [A, ..., [A, B]]] + ... \tag{33}$$

where

$$[Q_2, \Phi_3] = i\Phi_1, \quad [Q_2, [Q_2, \Phi_3]] = \Phi_3, \quad [Q_2, [Q_2, [Q_2, \Phi_3]]] = i\Phi_1 \tag{34}$$

We see that

$$\nu = \langle 0 \mid e^{i\frac{\pi}{2}Q_2}\Phi_1 e^{-i\frac{\pi}{2}Q_2} \mid 0\rangle \tag{35}$$

it is clear that it must be satisfied

$$Q_2 \mid 0\rangle \neq 0 \tag{36}$$

Similarly one shows that

$$Q_1 \mid 0\rangle \neq 0 \tag{37}$$

Notice that

$$e^{-i\frac{\pi}{2}T_3}\vec{\Phi} = \begin{pmatrix} -\Phi_2 \\ \Phi_1 \\ \Phi_3 \end{pmatrix} = e^{i\frac{\pi}{2}Q_3}\vec{\Phi}e^{-i\frac{\pi}{2}Q_3} \tag{38}$$

in particular

$$\begin{pmatrix} -Q_2 \\ Q_1 \\ Q_3 \end{pmatrix} = e^{i\frac{\pi}{2}Q_3}\begin{pmatrix} Q_1 \\ Q_2 \\ Q_3 \end{pmatrix}e^{-i\frac{\pi}{2}Q_3} \tag{39}$$

then we calculate

$$\begin{aligned} \langle 0 \mid [Q_2, \Phi_1] \mid 0\rangle &= \langle 0 \mid \left(-e^{i\frac{\pi}{2}Q_3}Q_1e^{-i\frac{\pi}{2}Q_3}e^{i\frac{\pi}{2}Q_3}\Phi_2e^{-i\frac{\pi}{2}Q_3}\right) \mid 0\rangle \\ &- \langle 0 \mid \left(-e^{i\frac{\pi}{2}Q_3}\Phi_2e^{-i\frac{\pi}{2}Q_3}e^{i\frac{\pi}{2}Q_3}Q_1e^{-i\frac{\pi}{2}Q_3}\right) \mid 0\rangle \qquad (40) \\ &= \langle 0 \mid \left(-e^{i\frac{\pi}{2}Q_3}Q_1\Phi_2e^{-i\frac{\pi}{2}Q_3} + e^{i\frac{\pi}{2}Q_3}\Phi_2Q_1e^{-i\frac{\pi}{2}Q_3}\right) \mid 0\rangle \\ &= -\langle 0 \mid [Q_1, \Phi_2] \mid 0\rangle \qquad (41) \end{aligned}$$

where we used $e^{\pm i\frac{\pi}{2}Q_3} \mid 0\rangle = \mid 0\rangle$, i.e. $Q_3 \mid 0\rangle = 0$ ($SO(2)$ remains unbroken).

Hence

$$\nu = \langle 0 \mid \Phi_3(0) \mid 0 \rangle = -\frac{i}{2} \left(\langle 0 \mid [Q_1, \Phi_2(0)] \mid 0 \rangle - \langle 0 \mid [Q_2, \Phi_1(0)] \mid 0 \rangle \right) \quad (42)$$

$$= -i \langle 0 \mid [Q_1, \Phi_2(0)] \mid 0 \rangle \quad (43)$$

$$= -i \int d^3x \langle 0 \mid [J_1^0(t,x), \Phi_2(0)] \mid 0 \rangle \quad (44)$$

Let us insert the identity operator

$$\sum\!\!\!\!\!\!\int \mid n\rangle\langle n \mid \tag{45}$$

Then we have

$$\begin{aligned}
\nu &= -i \int d^3x \langle 0 \mid [J_1^0(t,x), \Phi_2(0)] \mid 0\rangle && (46)\\
&= -i \sum\!\!\!\!\!\!\int \int d^3x \left(\langle 0 \mid J_1^0(t,x) \mid n\rangle\langle n \mid \Phi_2(0) \mid 0\rangle - \langle 0 \mid \Phi_2(0) \mid n\rangle\langle n \mid J_1^0(t,x) \mid 0\rangle \right) && (47)
\end{aligned}$$

Notice that

$$\langle n \mid J_1^0(t,x) \mid 0\rangle = \langle n \mid e^{i\mathcal{P}\cdot x} J_1^0(0) e^{-i\mathcal{P}\cdot x} \mid 0\rangle = e^{iE_n t - i\mathbf{p}_n\cdot\mathbf{x}} \langle n \mid J_1^0(0) \mid 0\rangle \tag{48}$$

Then we have

$$\nu = -i\sum\!\!\!\!\!\!\int d^3x \left(\langle 0 \mid J_1^0(0) \mid n\rangle\langle n \mid \Phi_2(0) \mid 0\rangle e^{-iE_n t + i\mathbf{p}_n\cdot\mathbf{x}} \right. \quad (49)$$

$$\left. -\langle 0 \mid \Phi_2(0) \mid n\rangle\langle n \mid J_1^0(0) \mid 0\rangle e^{iE_n t - i\mathbf{p}_n\cdot\mathbf{x}}\right) \quad (50)$$

one integrate over x hence

$$\nu = -i\sum\!\!\!\!\!\!\int \delta^{(3)}(\mathbf{p}_n) \left(\langle 0 \mid J_1^0(0) \mid n\rangle\langle n \mid \Phi_2(0) \mid 0\rangle e^{-iE_n t} \right. \quad (51)$$

$$\left. -\langle 0 \mid \Phi_2(0) \mid n\rangle\langle n \mid J_1^0(0) \mid 0\rangle e^{iE_n t}\right) \quad (52)$$

Then the sum is restricted to the states with $\mathbf{p}_n = 0$ momentum

$$\langle 0 \mid J_1^0(0) \mid n\rangle\langle n \mid \Phi_2(0) \mid 0\rangle = |c_n| e^{i\phi_n} \tag{53}$$

hence

$$\begin{aligned} \nu &= -i\sum\!\!\!\!\!\!\int \delta^{(3)}(\mathbf{p}_n)|c_n| \left(e^{i\phi_n} e^{-iE_n t} - e^{-i\phi_n} e^{iE_n t}\right) && (54) \\ &= 2\sum\!\!\!\!\!\!\int \delta^{(3)}(\mathbf{p}_n)|c_n| \sin(\phi_n - E_n t) && (55) \end{aligned}$$

It was silently assumed that Φ_k and J_n^0 are hermitian, but in Weinberg's famous textbook there is a proof without this constraint... one uses only properties of Lorentz invariance of the scalar field theory...

Conclusions

1. ν does not depend on time, hence $E_n = 0$
2. There must exist a state $\mid n\rangle$ for which $\langle n \mid \Phi_2(0) \mid 0\rangle \neq 0$ and $\langle n \mid J_1^0(0) \mid 0\rangle \neq 0$ but it is not a ground state because $\langle 0 \mid \Phi_{1,2}(0) \mid 0\rangle = 0$
3. There must exist states with $E_n = 0$ and $\mathbf{p}_n = 0$ which contribute to the sum, so these states are massless and are called Goldstone bosons

	Example	$O(N)$ linear sigma model	QCD
G	$O(3)$	$O(N)$	$SU(3)_L \times SU(3)_R$
n_G	3	$N(N-1)/2$	$16 = 8 + 8$
H	$O(2)$	$O(N-1)$	$SU(3)_V$
n_H	1	$(N-1)(N-2)/2$	8
# of Goldstones	2	$N-1$	8
Multiplet	$(\Phi_1(x), \Phi_2(x))$	$(\Phi_1(x), ..., \Phi_{N-1}(x))$	$i\bar{q}(x)\gamma_5\lambda_a q(x)$
Vacuum expectation	$v = \langle\Phi_3\rangle$	$v = \langle\Phi_N\rangle$	$v = \langle\bar{q}q\rangle$

Fermi Model for β-decay

$$n \to pe\overline{\nu}_e \tag{56}$$

Fermi proposed interaction Hamiltonian ($\mathcal{H}_F = -\mathcal{L}_F$)

$$\mathcal{L}_F = -\frac{G_F}{\sqrt{2}} \left(\overline{p}(x)\gamma_\lambda n(x)\right) \left(\overline{e}(x)\gamma^\lambda \nu(x)\right) + h.c. \tag{57}$$

where the Fermi constant

$$G_F \approx \frac{10^{-5}}{M_p^2} \tag{58}$$

Notice that this theory does not lead to the parity violation!

Fermi Model - modifications

- only left-handed neutrinos are observed!(e.g. **Pion decay**)
- the interaction currents have vector $-$ axial structure (Feynman and Gell-Mann)
- parity violation is maximal in the weak interaction (γ_5)
- lepton universality – the same coupling constant for all leptons and hadrons (almost Cabibbo angle)!

Left-Right handed neutrinos

- Left/Right-handed component of the spinor

$$\psi_{L/R} = \frac{1}{2}(1 \mp \gamma_5)\psi \tag{59}$$

Notice that

$$\psi = \psi_L + \psi_R \tag{60}$$

- $$\overline{\psi_L} = \frac{1}{2}\psi^\dagger(1-\gamma_5)\gamma_0 = \frac{1}{2}\psi^\dagger\gamma_0(1+\gamma_5) = \frac{1}{2}\overline{\psi}(1+\gamma_5) \tag{61}$$

- hence

$$\overline{\psi_L}\gamma_\mu\psi_R = \frac{1}{4}\overline{\psi}\left(1+\gamma_5\right)\gamma_\mu\left(1+\gamma_5\right)\psi = \frac{1}{4}\overline{\psi}\gamma_\mu\left(1-\gamma_5^2\right)\psi = 0 \tag{62}$$

but

$$\overline{\psi_L}\gamma_\mu\psi_L = \frac{1}{4}\overline{\psi}\left(1+\gamma_5\right)\gamma_\mu\left(1-\gamma_5\right)\psi = \frac{1}{2}\overline{\psi}\gamma_\mu\left(1-\gamma_5\right)\psi \neq 0 \tag{63}$$

- Then the current for weak interactions should have the form

$$\overline{\nu_{eL}}\gamma_\mu e = \overline{\nu_{eL}}\gamma_\mu e_L \tag{64}$$

- for the hadronic current also only left-handed fields!

Dirac mass term

- Hence

$$\mathcal{L}_{Fermion} = \overline{\psi}(i\gamma_\mu\partial^\mu - M)\psi \quad (65)$$

$$= i\overline{\psi_L}\partial^\mu\gamma_\mu\psi_L + i\overline{\psi_R}\partial^\mu\gamma_\mu\psi_R - M\overline{\psi_R}\psi_L - M\overline{\psi_L}\psi_R \quad (66)$$

* **Lack of right-handed neutrinos means that $M = 0$?!**
  - Notice that

$$(\overline{\psi_R}\psi_L)^\dagger = \frac{1}{4}\psi^\dagger(1-\gamma_5)\gamma_0(1+\gamma_5)\psi = \overline{\psi_L}\psi_R \quad (67)$$

Modified Fermi Model

- $$\mathcal{L}_{eff} = -\frac{G_F}{\sqrt{2}} \mathcal{J}^{\dagger}_{\lambda}(x) \mathcal{J}^{\lambda}(x) + h.c. \tag{68}$$
- $$\mathcal{J}^{\lambda}(x) = \mathcal{J}^{\lambda}_{lep}(x) + \mathcal{J}^{\lambda}_{had}(x) \tag{69}$$
- $$\begin{aligned} \mathcal{J}^{\lambda}_{lep}(x) &= \overline{\nu}_e(x)\gamma^{\lambda}(1-\gamma_5)e(x) + \overline{\nu}_{\mu}(x)\gamma^{\lambda}(1-\gamma_5)\mu(x) && (70) \\ \mathcal{J}^{\lambda}_{had}(x) &= \overline{p}(x)\gamma^{\lambda}(1-\gamma_5)n(x) && (71) \end{aligned}$$
- Notice that
$$\overline{\psi}\gamma^{\lambda}(1-\gamma_5)\phi = 2\overline{\psi_L}\gamma^{\lambda}\phi_L \tag{72}$$

Problems of Fermi Model: Unitarity

- At the tree level the Fermi model is not unitary
- consider

$$\nu_e + e \to \nu_e + e \tag{73}$$

- in CMF

$$\frac{d\sigma}{d\Omega} = G_F^2 \frac{k^2}{\pi^2}, \quad \sigma = \frac{4G_F^2 k^2}{\pi} \tag{74}$$

  k is momentum in CMF
- Point-like spin $1/2$ particles interaction is described by s-wave partial wave, then from unitarity one gets

$$\sigma < \frac{\pi}{2k^2} \tag{75}$$

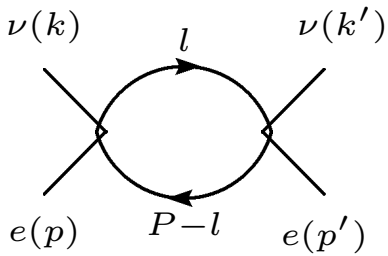
- Above

$$k_{lim}^4 = \frac{\pi^2}{8G_F^2} \quad k_{lim} \approx 300\ GeV$$

  the unitarity is broken!
- Higher order terms needed!

Problems of Fermi Model: Renormalizabilty

- Model is not renormalizable
  - $G_F \sim 1/M_p^2$, thus the higher order contributions are increasingly divergent
  - moreover, the loop integrals diverge:



$$\sim \int d^4 l \frac{l_\alpha \gamma^\alpha + m_e}{l^2 - m_e^2} \gamma_\mu (1-\gamma_5) \frac{(P-l)_\lambda \gamma^\lambda}{(P-l)^2} \gamma^\mu (1-\gamma_5) \quad (76)$$

$$\sim \int dl \frac{l^3\, l^2}{l^4} \sim \int dl\; l \to \infty \quad (77)$$

where $P = k + p$.

$$(P-l)^2 = s + l^2 - 2P \cdot l$$

- We need vector intermediate bosons!

Intermediate Vector Bosons (IVB) Theory

IVB Model

$$\mathcal{L}_{IVB} = g\left(J_\mu W^\mu + h.c.\right) \tag{78}$$

- the problem with lack of unitarity remains but visible at higher energies...
- g is small and dimensionless but
- the theory is still not renormalizable...
- Need for better theory

Choice of the SU(2)xU(1) group: the algebraic approach

Weak (charged-current) interactions and vector bosons

$$\mathcal{L}_W = g\left(J^+_\lambda W^\lambda + h.c.\right) \tag{79}$$

the lepton weak current:

$$J^+_\lambda = \bar{\nu}_e \gamma_\lambda (1-\gamma_5) e \tag{80}$$

where $J^- = (J^+)^\dagger$

Electromagnetic interactions and photon

$$\mathcal{L}_{em} = e J^{em}_\lambda A^\lambda \tag{81}$$

the electromagnetic current:

$$J^{em}_\lambda = -\bar{e}\gamma_\lambda e \tag{82}$$

Weak currents have $T_\pm$ $SU(2)$ generator form!

Idea: $J^\pm_\lambda$ and J^{em}_λ generate an algebra?

Charged current Weak and Electric Currents

$$T_+(t) = \frac{1}{2}\int d^3x J_0^+(t,\mathbf{x}) = \frac{1}{2}\int d^3x \bar{\nu}_e(t,\mathbf{x})\gamma_0(1-\gamma_5)e(t,\mathbf{x}) \quad (83)$$

$$= \frac{1}{2}\int d^3x \nu_e^\dagger(t,\mathbf{x})(1-\gamma_5)e(t,\mathbf{x}) \quad (84)$$

$$T_-(t) = (T_+)^\dagger \quad (85)$$

From

$$[T_+, T_-] = 2T_3 \quad (86)$$

we have

$$T_3 = \frac{1}{4}\int d^3x [\nu_e^\dagger(1-\gamma_5)\nu_e - e^\dagger(1-\gamma_5)e] \quad (87)$$

$$Q(t) = \int d^3x J_0^{em}(t,\mathbf{x}) = -\int d^3x \bar{e}(t,\mathbf{x})\gamma_0 e(t,\mathbf{x}) = -\int d^3x e^\dagger(t,\mathbf{x})e(t,\mathbf{x}) \quad (88)$$

we see that $T_3 \neq Q$ but also $[Q, T_i] \neq 0$

$$\{\psi_i(t,\mathbf{x}), \psi_j^\dagger(t,\mathbf{y})\} = \delta_{ij}\delta^{(3)}(\mathbf{x}-\mathbf{y}) \quad (89)$$

There are two alternatives

i) one can introduce T_3 as additional interaction current in the theory and form $SU(2) \times U(1)$

ii) One can add new fermion to the multiplet and use $T_{\pm}$ and Q to form the $SU(2)$ – rejected by experiments

Notice that $\Psi^\dagger(1-\gamma_5)\psi = \frac{1}{2}\Psi^\dagger(1-\gamma_5)(1-\gamma_5)\psi = 2\Psi_L^\dagger\psi_L$

$$T_+ = \int d^3x \nu_{eL}^\dagger e_L \tag{90}$$

$$T_3 = \frac{1}{2}\int d^3x [\nu_{eL}^\dagger \nu_{eL} - e_L^\dagger e_L] \tag{91}$$

$$Q = -\int d^3x e^\dagger e = -\int d^3x (e_L^\dagger e_L + e_R^\dagger e_R) \tag{92}$$

On the other hand it can be shown that

$$Q - T_3 = \int d^3x \left[-\frac{1}{2}(\nu_{eL}^\dagger \nu_{eL} + e_L^\dagger e_L) - e_R^\dagger e_R\right] \tag{93}$$

It can be shown that

$$[Q - T_3, T_i] = 0 \tag{94}$$

Hence we have two symmetry groups generated by $Q - T_3$ and T_is. With respect to $SU(2)$ the right-handed fields are singlets, while the left-handed fields form doublets!

We may consider left-handed doublets and right-handed singlets (quarks and charged leptons) — the right-handed fields are needed to build the fermion mass terms!

$$\begin{pmatrix} \nu_{e,L} \\ e_L \end{pmatrix}, \quad \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad e_R, \quad u_R, \quad d_R \tag{95}$$

$$T_+ = \int d^3x\,(\nu_{eL}^\dagger e_L + u_L^\dagger d_L) \tag{96}$$

$$T_+^\dagger = T_- \tag{97}$$

$$T_3 = \frac{1}{2}\int d^3x(\nu_{eL}^\dagger \nu_{eL} - e_L^\dagger e_L + u_L^\dagger u_L - d_L^\dagger d_L) \tag{98}$$

$$Q = \int d^3x\left(-e^\dagger e + \frac{2}{3}u^\dagger u - \frac{1}{3}d^\dagger d\right) \tag{99}$$

$$= \int d^3x\left(-e_L^\dagger e_L - e_R^\dagger e_R + \frac{2}{3}u_L^\dagger u_L + \frac{2}{3}u_R^\dagger u_R - \frac{1}{3}d_L^\dagger d_L - \frac{1}{3}d_R^\dagger d_R\right) \tag{100}$$

$$Q - T_3 = \int d^3x \left[-\frac{1}{2}(\nu_{eL}^\dagger \nu_{eL} + e_L^\dagger e_L) + \frac{1}{6}(u_L^\dagger u_L + d_L^\dagger d_L) - e_R^\dagger e_R + \frac{2}{3} u_R^\dagger u_R - \frac{1}{3} d_R^\dagger d_R \right] \quad (101)$$

Hypercharge operator

$$Y = 2(Q - T_3) \quad (102)$$

$$Y = \int d^3x \left[-(\nu_{eL}^\dagger \nu_{eL} + e_L^\dagger e_L) + \frac{1}{3}(u_L^\dagger u_L + d_L^\dagger d_L) - 2e_R^\dagger e_R + \frac{4}{3} u_R^\dagger u_R - \frac{2}{3} d_R^\dagger d_R \right] \quad (103)$$

$T_\pm$ and T_3 and Y are generators of the group

$$SU_L(2) \times U_Y(1) \quad (104)$$

Final Quantum Numbers

$SU(2)_L$

Generated by $T_\pm$ and T_3.
Fermion doublets:

$$l_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \quad q_L = \begin{pmatrix} u \\ d \end{pmatrix}_L \tag{105}$$

Fermion singlets:

$$e_R,\ u_R,\ d_R, \tag{106}$$

$U(1)_Y$

Generated by Y.

$$Y(l_L) = -1,\ Y(q_L) = 1/3,\ Y(e_R) = -2,\ Y(u_R) = 4/3,\ Y(d_R) = -2/3 \tag{107}$$

Y is twice the average charges of each multiplet, because T_3 average value within multiplet is always zero!!!